

Computer Organization and Architecture 8085 Instructions Set

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8085 Instruction Set:

1. Data Transfer Instructions
2. Arithmetic Instructions
3. Logical Instructions
4. Branching Instructions
5. Control Instructions

Simulator: <https://learning-microprocessors.sourceforge.io/8085-simulator.html>

Starting the Simulator:

```
C:\Users\CEDDU\Downloads>java -jar 8085Compiler.jar
```

PSW Instruction in 8085:

- The PSW instruction in the 8085 microprocessor is used to read and write the Program Status Word (PSW).
- The PSW is a 16-bit register that stores the status of the processor after an instruction is executed.
- It is used to store flags that reflect the result of the most recent arithmetic or logical operations.
- After performing operations like addition or subtraction, you can use the flags in the PSW to check if there was a carry, if the result was zero, if the number was negative, etc.

How to use the PSW instruction ?

- The PUSH PSW instruction copies the contents of the PSW to the stack.
- The POP PSW instruction copies the contents of the top two memory locations of the stack to the PSW.

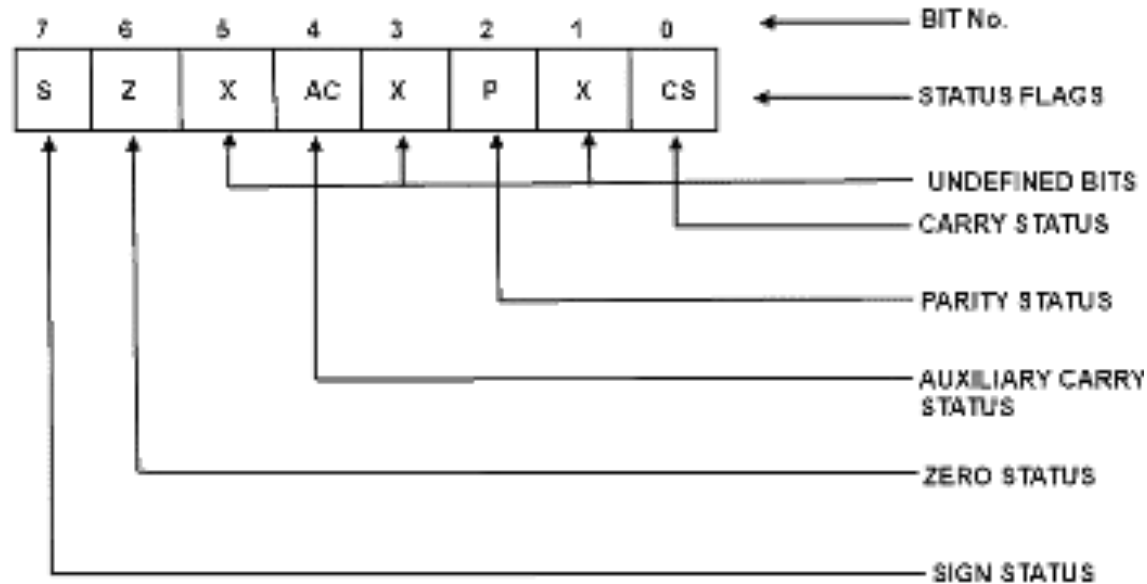


Figure 2: Status Flags of Intel 8085

CALL and RET Instructions:

- In the 8085 microprocessor, the **CALL** and **RET** instructions are used to implement **subroutine calls**.
- A subroutine is a block of code that performs a specific task and can be executed multiple times from different places in the program.
- The **CALL** instruction is used to jump to a subroutine, while the **RET** instruction is used to return from the subroutine to the point where it was called.
- **SYNTAX for Unconditional Call:**

CALL <16-Bit Addr or Label>

Difference between JMP and CALL:

The main differences between a JMP instruction and a CALL instruction is as follows –

- If a JMP instruction is executed, we jump to the destination location, and the execution carries on from there, without bothering to come back later to the instruction after the JMP. On the other hand, if a CALL instruction is executed, we jump to the subroutine, and the execution carries on from there till the RET instruction is executed in the subroutine, and then we come back to the instruction after the CALL in the main program.
- The address of the next instruction after the CALL instruction is called the return address. This is the address to which the program flow returns when the RET instruction is executed by the 8085. By the time the 8085 fetches the call instruction, the PC would have been incremented by 3, and will be pointing to the next instruction after call. In other words, PC will have the return address by the time the call instruction is fetched.
- In order to facilitate such a return, the CALL instruction will first of all store above the top of stack, the return address. Only then the branch to the subroutine takes place.

Conditional Calls and Returns:

CC, CNC, CP, CM, CPE, CPO, CZ, CNZ

RC, RNC, RP, RM, RPE, RPO, RZ

CC: Call if carry flag is 1

CNC: Call if carry flag is 0

CP: Calls if sign flag is 0

CM: Calls if sign flag is 1

CPE: Calls if parity flag is 1

CPO: Calls if parity flag is 0

CZ: Calls if zero flag is 1

CNZ: Calls if zero flag is 0

Stack Operations:

PUSH Operation:

PUSH Rp

This is a 1-byte instruction. This instruction copies the contents of the specified register pair on the stack as described below:

- The stack pointer is decremented and the contents of the higher-order register are copied to the location shown by the stack pointer register.
- The stack pointer is again decremented and the contents of the low-order register are copied to that location.

POP Rp

This is a 1-byte instruction. This instruction copies the contents of the top two memory locations of the stack into the specified register pair.

- First, the contents of the memory location indicated by the stack pointer register are copied into the low-order register and then the stack pointer register is incremented by 1.
- The contents of the next memory location are copied into the high-order register and the stack pointer register is again incremented by 1.

EXAMPLE:

Registers

A/PSW	0x 00 02
BC	0x 00 00
DE	0x 15 23
HL	0x 00 00
SP	0x 00 00
PC	0x 00 00

main.asm

```
1 PUSH D
2 HLT
```

Registers

A/PSW	0x 00 02
BC	0x 00 00
DE	0x 15 23
HL	0x 00 00
SP	0x FF FD
PC	0x 08 02

main.asm

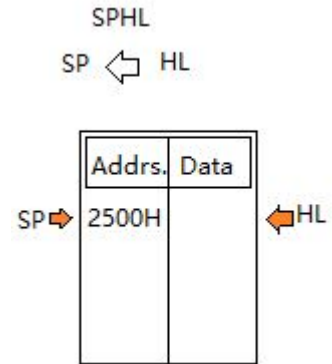
```
1 PUSH D
2 POP H
3 HLT
```

Registers

A/PSW	0x 00 02
BC	0x 00 00
DE	0x 15 23
HL	0x 15 23 
SP	0x FF FF
PC	0x 08 03

SPHL Instruction:

- **SPHL Instruction (Set Stack Pointer to HL Register Pair).**
- The SPHL instruction in the 8085 microprocessor copies the contents of the HL register pair to the stack pointer (SP).
- The **HL register pair** holds a 16-bit address, which is stored in two 8-bit registers: **H** (high byte) and **L** (low byte).
- The **SPHL** instruction copies the 16-bit value from the HL register pair into the **stack pointer (SP)**, effectively changing the location of the stack to the address specified by HL.
- This is useful for changing the stack's location dynamically during program execution.



XTHL Instruction (Exchange Top of Stack with HL Register Pair):

- The XTHL instruction in the 8085 microprocessor exchanges the contents of the H and L registers with the top of the stack.
- The **XTHL** instruction exchanges the value located at the address in **SP** with the value in the **HL register pair**.
- Specifically, the **L** register's value is exchanged with the **memory content at the address pointed by SP**, and the **H** register's value is exchanged with the **memory content at the address pointed by SP+1**.
- After this exchange, the **SP** is incremented by 2, as the top of the stack (2 bytes) has been swapped.

PCHL Instruction (Program Counter to HL Register Pair):

- The PCHL instruction is used to load the program counter with the contents of registers H and L.
- The PCHL instruction in the 8085 microprocessor copies the contents of registers H and L into the program counter. The contents of H are placed as the high-order byte and the contents of L as the low order byte.



Thanks

